

Building Effective Enterprise Mobile Applications

Enterprise Mobile Architecture

What is Mobile Software?

● What is Mobile?

- Away from the desk out of the office
- On the road traveling
- On the factory floor – always moving
- Engaging face to face with your customers

● What is Mobile Software?

- A device in your hands that helps you with your process
- E.g. Capturing data, looking up data
- Doesn't mean wireless!
- Concept of offline/online

Why Mobile Software?

- **Why develop mobile software?**
 - Information/data at your fingertips
 - Make decisions on the spot
 - Perform your process on the spot
- **What make mobile software “Enterprise” mobile software**
 - Access and integration to enterprise information and software assets
 - E.g. Lookup: Customer, Product and Stock information
 - E.g. Capture Data: New Customer Order, Customer Visit information

The Benefits of Mobile Software

- **Reduce paper costs**
- **Reduce communication costs**
- **Processes automation**
- **Increase Productivity**
- **Increase Service turn around**
- **It is fundamentally about:**
 - **Saving costs**
 - **Making more money**
 - **Capturing data more efficiently**
 - **Providing data back more efficiently**

Usage Examples

- **Couriers**

- **Dispatch, Pickup and Sign**

- **Territory Managers**

- **View products/stock, Place orders**

- **Auditors/Insurance**

- **Complete audits, print report & quotes**

- **Tourists**

- **Why can I find information about places in this city?**



Some Thoughts on Design Goals for Mobile

- **What problem(s) are you trying to solve?**
- **What system(s) are involved?**
- **What data do you need from these systems?**
- **How will you secure your mobile system?**
- **How will you connect your mobile users?**

What is Mobile Software?

Architectures
Real Time vs. Synchronized

Telecommunications

Devices

System Integration

Why is Mobile Software Unique?

- You can't always trust the mobile network
- Devices have limited foot prints
- Languages and development environments are often proprietary
- But
 - You still need good software design
 - You still need good software architectures

The Spectrum of Architectures

- **In mobile it's all about transactions**
- **But how do you do these transactions?**
- **Real Time**
 - **Real time over a permanent network**
 - **Process Data in Real time**
 - **Otherwise known as “Always Connected”**
- **Synchronised**
 - **Process and store data on the device**
 - **Synchronise when needed**
 - **Otherwise know as “Cached”**

Real Time Architectures

- **Pros**

- Information is always up to date
- Data gets processed in the backend faster

- **Cons**

- Poorer user experience on low bandwidth networks.
- Doesn't cope well with large amounts of data

- **Working Examples**

- Warehouse Management Systems
- High Bandwidth / Discrete Transactions

Synchronized Architectures

● Pros

- Good for dealing with unreliable networks
- Data is either synchronised or not
- Only changes are synchronised
- User experience is much faster
- User does not need to be online to process

● Cons

- Data can become stale if synchronisation does not often enough
- Data is pumped into backend system “batch based”

● Working Examples

- Sales Force applications
- Route Management applications
- Anything with large amounts of reference data

Types of Mobile Applications

● Smart Clients

- Run locally on the device
- Uses local processing power
- Uses local resources
- UI rendered on the device
- Real time / Synchronized applications
- Key focus for Enterprise Mobile Software

● Mobile Web Applications

- Server Side execution – HTML, WAP etc
- Renders to specific device capability
- UI rendered on the server
- Real time applications only

Types of Mobile Application Continued

What about writing for multiple devices?

Multi-device applications issues

- Different form factors
- Different device capabilities
- Different browser standards

Cross platform issues

- Different platforms
- Different APIs
- Different languages

Mobile Web Pages

Local Code

Mobile Web
Browser

Device Operating System



Mobile Application Devices

- **Low end mobile phones & pagers**
- **Smart Devices**
- **Full Computer Experience**
- **Device separated according to**
 - **Processor Ability**
 - **Memory / Storage Capacity**
 - **User input methods**

Mobile Application Devices

Low End - Mobile Web Experience

- Mobile Phones and Pagers
- Micro browsers
 - WAP, HTML, XML, cHTML
- Key based user input
- Many different devices



Mobile Application Devices

Smart Devices

- Application on Device
- Stylus Input
- Online/Offline Capability
- Local processing power



Mobile Application Devices

Full PC Experience

- Full processing power
- Full screen display
- Tablet PC / Laptop



Telecommunications

- **Build your architectures to support TCP/IP**
 - **And you can use any of these technologies!**
- **But test, test, test!**
 - **Synchronized architectures work better over slower connections**
 - **Real time architectures tend to need faster connections for good user experience.**
 - **Always check for timeouts**
 - **Applications need to handle “no network” gracefully**

System Integration

- **All this data is great but what about?**
 - **Integrating to / from existing systems?**
- **Where does the data go?**
- **Where do you need to get data from?**
- **Examples**
 - **Where is your master customer list?**
 - **Where do your mobile orders get processed?**

System Integration

- **Fully integrated vs. Print-outs**
- **System Integration can make or break a mobile application**
- **Concept of a central mobile database**
 - **Merge all information required for devices into one place**
 - **Consolidate all mobile data captured into one place and integrate back in one go**
- **Shouldn't matter whether synchronized or real time architecture**

Design Issues

- **User Interface is critical**
 - For user adoption
 - Must be simple yet powerful
- **Data resilience is imperative**
 - Synchronisation must be seamless
 - It either works or it doesn't with nothing in between
- **Data partitioning simplifies overheads**
 - Deliver data to the users that need it

Design Issues cont.

- **Data partitioning**
 - e.g. don't get every sales person every other sales person's clients
 - This is horizontal partitioning.
 - Each mobile device has only the data it needs
 - Static (All) vs. Static (Personal)
 - Vertical partitioning is about only pushing the data types on need on a devices
 - Always determine how much data you need and not more.

Scalability

- **To support thousands of users you must consider**
 - **Network / telecoms usage for all your users**
 - **Multiple synchronisation points**
 - **Over the air update frameworks**
 - **Data storage capacity**
 - **Keep your audit logs small**



Seamless Deployment

- **Seamless Synchronisation**
 - One Button
 - Incorporate over-the-air updates
- **Removable Media Card Install**
 - e.g. Secure Digital, Compact Flash
 - Lower network costs
- **Enterprise Device Deployment Tool**

Data Integrity

- **Data Synchronisation**
 - **Initialisation**
 - **Delta based updates**
 - **A synchronisation is transactional**
 - **Either works or not!**
- **Test power failure scenarios**
- **Reinitialise data over the air if required**

What do we need? - Concepts

- **Online/Offline Capability**
 - **Offline – Replicated Data (Main Function)**
 - **Online – Real time Lookups**
- **Security**
 - **Security access through firewall tunnel**
- **Scalability**
 - **The ability to scale to 1000s of users**
- **Integrated Approach**
 - **Consolidate data to/from Backend Systems**

What do we need? - Layers

- **User Interface Layer**
- **Telecommunications Layer**
- **Application / Web Server**
- **Data Repository**
- **Integration Layer**
- **Backend Systems**

Physical Architecture

